

Project Title: *(Students to define specific focus)*

Authors

Dr. Idris Abdulmumin
Prof Vukosi Marivate

High-Level Topic:

Automatic Detection of Machine Generated Texts

Background and Motivation:

Africa is home to over 2,000 languages from more than six language families and has the highest linguistic diversity among all continents. These include 75 languages with at least one million speakers each. Yet, there is little NLP research conducted on African languages. Crucial to enabling such research is the availability of high-quality annotated datasets.

The rapid advancement of large language models (LLMs) has enabled them to excel in various tasks, including content generation. However, their unregulated use poses risks, such as plagiarism, particularly in low-resource languages. Existing machine-generated text detectors are primarily trained on high-resource languages like English and French, leaving African languages underserved. This project aims to develop automatic detectors capable of distinguishing between human- and machine-generated text in African languages. Human-generated texts can be sourced from datasets like Common Crawl or other existing texts, while machine-generated texts can be obtained by prompting LLMs such as GPT, Gemini, Gemma, and Llama.

Research Questions & Objectives:

Students should define their own specific research questions, but they may consider:

1. Which transfer learning approaches (e.g., cross-lingual models, adapter-based methods) best improve machine-generated text detection in African languages?
2. What evaluation techniques can help analyze errors and biases in machine-generated text detection for African languages?
3. How do different LLMs generate text in African languages, and what patterns emerge in their outputs compared to human-written text?
4. What linguistic and stylistic features are most effective for detecting machine-generated text in low-resource African languages?

Dataset & Resources:

- **Various Datasets in African Languages:** E.g., Mafand-MT, Vukuzenzele, AfriSenti, Commoncrawl extracts, etc.
- Pre-trained multilingual models (e.g., mBERT, AfriBERTa, AfroXLMR, XLM-Roberta).
- Additional NLP toolkits for error analysis.

Methodology:

Students should explore and justify their choices in methodology. Possible approaches include:

- **Baseline Models:** Train and evaluate standard classification models for machine- and human-generated texts.
- **Transfer Learning:** Experiment with fine-tuning multilingual models or adapter-based techniques.
- **Data Augmentation:** Use techniques like back-translation, adversarial data augmentation, or synthetic data generation.
- **Explanatory Evaluation:** Conduct error analysis using SHAP, LIME, or attention visualization to understand misclassifications.

Languages to Consider:

Students should choose a subset of languages based on linguistic similarities, availability of data, or societal impact.

Evaluation Metrics:

Students must define a comprehensive evaluation strategy beyond standard classification metrics. Consider:

- **Performance Metrics:** F1-score, Precision, Recall, Accuracy.
- **Robustness Metrics:** Performance across different dialects or domains.
- **Explanatory Analysis:** Model interpretation methods to identify patterns of failure and linguistic challenges.

Expected Deliverables:

- **Proposal:** A concise research proposal outlining the approach.
- **Implementation & Experiments:** A well-documented code repository.
- **Report:** A final report
- **Presentation:** A summary of findings.

References & Further Reading:

Students should explore relevant papers, blogs, and tutorials on text classification, transfer learning, and model interpretability.

- **AfriHate: A Multilingual Collection of Hate Speech and Abusive Language Datasets for African Languages** - [Paper](#) | [Dataset](#)
 - **BRIGHTER: BRIdging the Gap in Human-Annotated Textual Emotion Recognition Datasets for 28 Languages** - [Paper](#) | [Dataset](#) | [Shared Task](#)
 - **EthioEmo: Evaluating the Capabilities of Large Language Models for Multi-label Emotion Understanding** - [Paper](#) | [Dataset](#)
 - **AfriSenti: A Twitter Sentiment Analysis Benchmark for African Languages** - [Paper](#) | [Dataset](#)
 - **SemEval-2023 Task 12: Sentiment Analysis for African Languages (AfriSenti-SemEval)** - [Paper](#)
 - **A Sentiment Corpus for South African Under-Resourced Languages in a Multilingual Context** - [Paper](#) | [Dataset](#) (Shona)
 - **A Sesotho News Headlines Dataset for Sentiment Analysis** - [Paper](#) | [Dataset](#)
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